

Portfolio Performance Evaluation

Introduction

- If markets are efficient, investors must be able to measure asset management performance
- Two common ways to measure average portfolio return:
 1. Time-weighted returns
 2. Dollar-weighted returns
- Returns must be adjusted for risk.

Dollar- and Time-Weighted Returns

Time-weighted returns

- The **geometric average** is a time-weighted average.
- Each period's return has equal weight.

$$(1 + r_G)^n = (1 + r_1)(1 + r_2) \dots (1 + r_n)$$

Dollar- and Time-Weighted Returns

Dollar-weighted returns

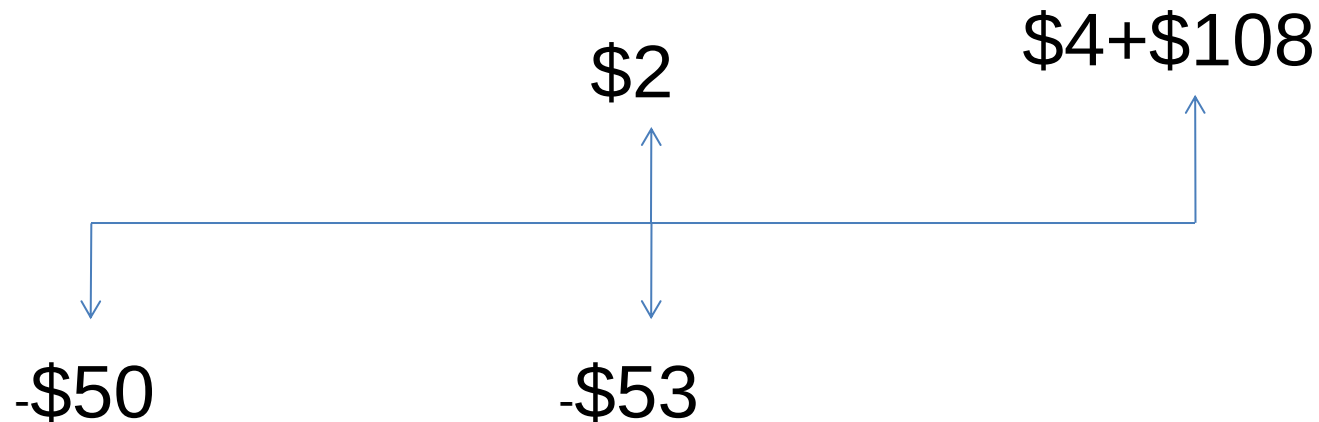
- Internal rate of return considering the cash flow from or to investment
- Returns are weighted by the amount invested in each period:

$$PV = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \dots \frac{C_n}{(1+r)^n}$$

Example of Multiperiod Returns

Time	Outlay
0	\$50 to purchase first share
1	\$53 to purchase second share a year later
Proceeds	
1	\$2 dividend from initially purchased share
2	\$4 dividend from the 2 shares held in the second year, plus \$108 received from selling both shares at \$54 each

Dollar-Weighted Return



Dollar-weighted Return (IRR):

$$-50 = \frac{-51}{(1+r)^1} + \frac{112}{(1+r)^2}$$

$$r = 7.117\%$$

Time-Weighted Return

$$r_1 = \frac{53 - 50 + 2}{50} = 10\%$$

$$r_2 = \frac{54 - 53 + 2}{53} = 5.66\%$$

$$r_G = [(1.1) (1.0566)]^{1/2} - 1 = 7.81\%$$

The dollar-weighted average is less than the time-weighted average in this example because more money is invested in year two, when the return was lower.

Adjusting Returns for Risk

- The simplest and most popular way to adjust returns for risk is to compare the portfolio's return with the returns on a comparison universe.
- The comparison universe is a benchmark composed of a group of funds or portfolios with similar risk characteristics, such as growth stock funds or high-yield bond funds.

Risk Adjusted Performance: Sharpe

1) Sharpe Index

$$\frac{(\overline{r_P} - \overline{r_f})}{\sigma_P}$$

$\overline{r_p}$ = Average return on the portfolio

$\overline{r_f}$ = Average risk free rate

σ_p = Standard deviation of portfolio
return

Risk Adjusted Performance: Treynor

2) Treynor Measure

$$\frac{(\overline{r_P} - \overline{r_f})}{\beta_P}$$

$\overline{r_P}$ = Average return on the portfolio

$\overline{r_f}$ = Average risk free rate

β_P = Weighted average beta for portfolio

Risk Adjusted Performance: Jensen

3) Jensen's Measure

$$\alpha_p = \bar{r}_p - \left[\bar{r}_f + \beta_p (\bar{r}_M - \bar{r}_f) \right]$$

α_p = Alpha for the portfolio

$\bar{r}_p$ = Average return on the portfolio

β_p = Weighted average Beta

$\bar{r}_f$ = Average risk free rate

$\bar{r}_m$ = Average return on market index portfolio

Information Ratio

$$\text{Information Ratio} = \alpha_p / \sigma(e_p)$$

The information ratio divides the alpha of the portfolio by the nonsystematic risk.

Nonsystematic risk could, in theory, be eliminated by diversification.

Morningstar Risk-Adjusted Return

$$MRAR(\gamma) = \left[\frac{1}{T} \sum_{t=1}^T \left(\frac{1 + r_t}{1 + r_{ft}} \right)^{-\gamma} \right]^{\frac{12}{\gamma}} - 1$$

MRAR -> , 0

= (,)

: ?

sharp ratio MRAR -> MRAR?

- Uses average excess monthly returns
- γ measures risk aversion
- Can be thought of as the risk-free equivalent excess return of P with a certain risk aversion (γ)

M^2 Measure

- Developed by **Modigliani and Modigliani**
- Create an adjusted portfolio (P^*) that has the same standard deviation as the market index.
- Because the market index and P^* have the same standard deviation, their returns are comparable:

$$M^2 = r_{P^*} - r_M$$

M^2 Measure: Example

Managed Portfolio: return = 35% standard deviation = 42%

Market Portfolio: return = 28% standard deviation = 30%

T-bill return = 6%

P* Portfolio:

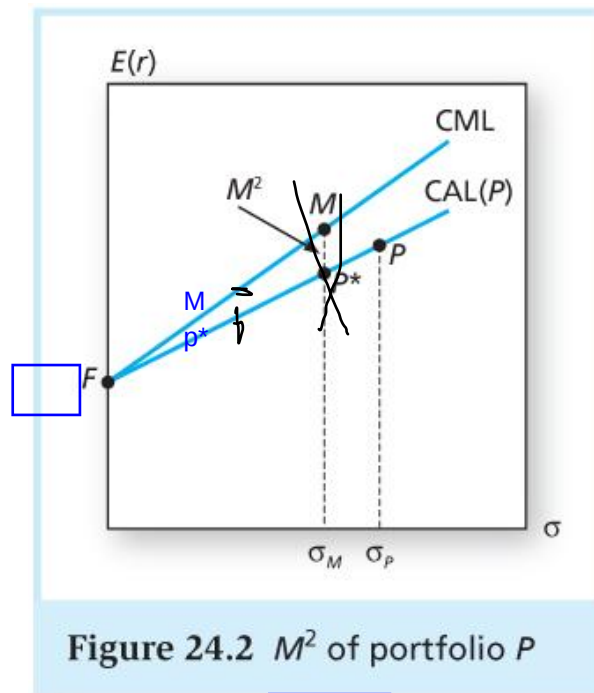
30% ?

$30/42 = .714$ in P and $(1-.714)$ or $.286$ in T-bills

The return on P* is $(.714) (.35) + (.286) (.06) = 26.7\%$

Since this return is less than the market, the managed portfolio underperformed.

M^2 of Portfolio P



Which Measure is Appropriate?

It depends on investment assumptions

- 1) If P is not diversified, then use the Sharpe measure as it measures reward to risk. X -> sharp ratio
- 2) If the P is diversified, non-systematic risk is negligible and the appropriate metric is Treynor's, measuring excess return to beta.

Portfolio Performance

	Portfolio P	Portfolio Q	Market
Beta	.90	1.60	1.0
Excess return ($\bar{r} - \bar{r}_f$)	11%	19%	10%
Alpha*	2%	3%	0

Table 24.1

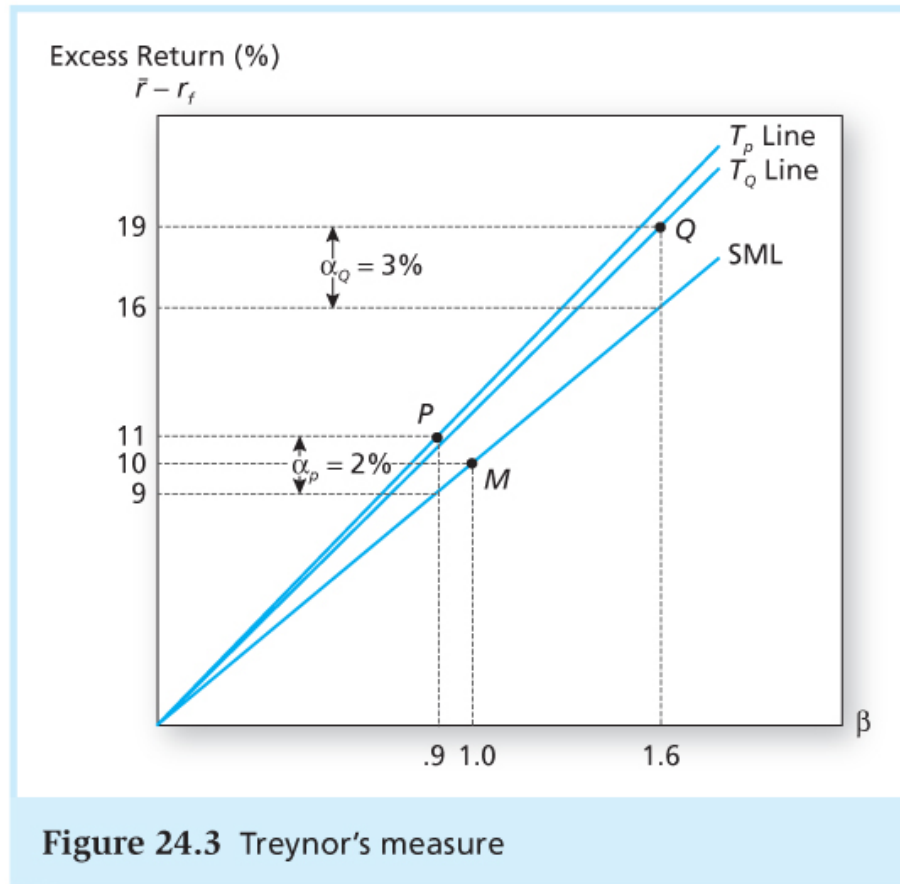
Portfolio performance

*Alpha = Excess return – (Beta × Market excess return)

$$= (\bar{r} - \bar{r}_f) - \beta(\bar{r}_M - \bar{r}_f) = \bar{r} - [\bar{r}_f + \beta(\bar{r}_M - \bar{r}_f)]$$

Is Q better than P?

Treynor's Measure



Performance Statistics

Table 24.3

Performance statistics

	Portfolio P	Portfolio Q	Portfolio M
Sharpe ratio	0.43	0.49	0.19
M^2	2.16	2.66	0.00
Morningstar RAR	0.30	0.80	0.07
SCL regression statistics			
Alpha	1.63	5.26	0.00
Beta	0.70	1.40	1.00
Treynor	3.97	5.38	1.64
T^2	2.34	3.74	0.00
$\sigma(e)$	2.02	9.81	0.00
Information ratio	0.81	0.54	0.00
R-SQR	0.91	0.64	1.00

Interpretation of Table 24.3

- If P or Q represents the entire investment, Q is better because of its higher Sharpe measure and better M^2 .
- If P and Q are competing for a role as one of a number of subportfolios, Q also dominates because its Treynor measure is higher.
- If we seek an active portfolio to mix with an index portfolio, **P is better due to its higher information ratio.**

Performance Measurement for Hedge Funds

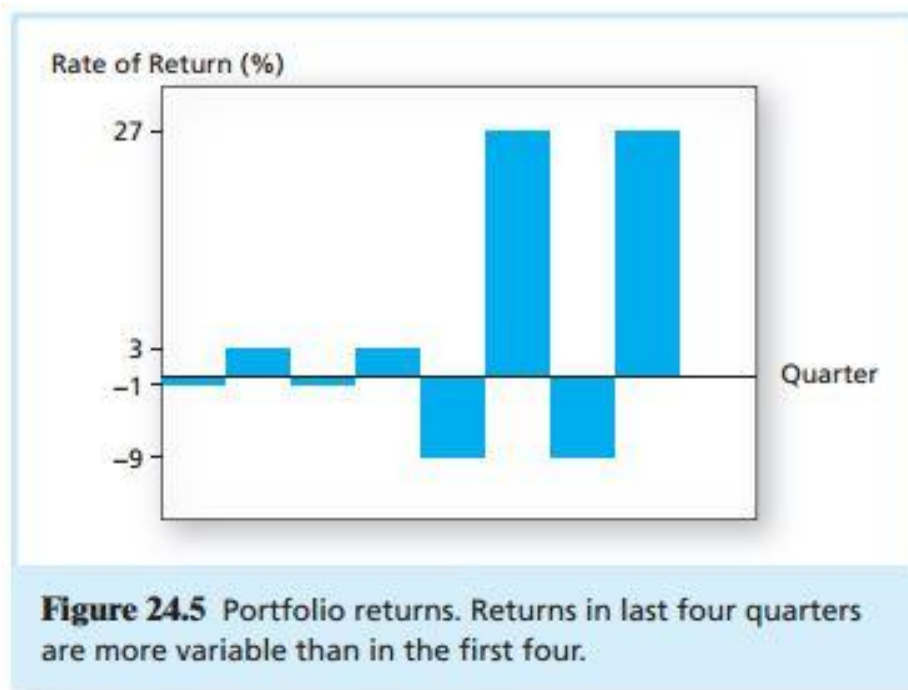
- When the hedge fund is optimally combined with the baseline portfolio, the improvement in the Sharpe measure will be determined by its information ratio:

$$S_P^2 = S_M^2 + \left[\frac{\alpha_H}{\alpha(e_H)} \right]^2$$

Performance Measurement with Changing Portfolio Composition

- We need a very long observation period to measure performance with any precision, even if the return distribution is stable with a constant mean and variance.
- What if the mean and variance are not constant? We need to keep track of portfolio changes.

Portfolio Returns



Market Timing

- In its pure form, market timing involves **shifting funds** between a market-index portfolio and a safe asset.

- Treynor and Mazuy:

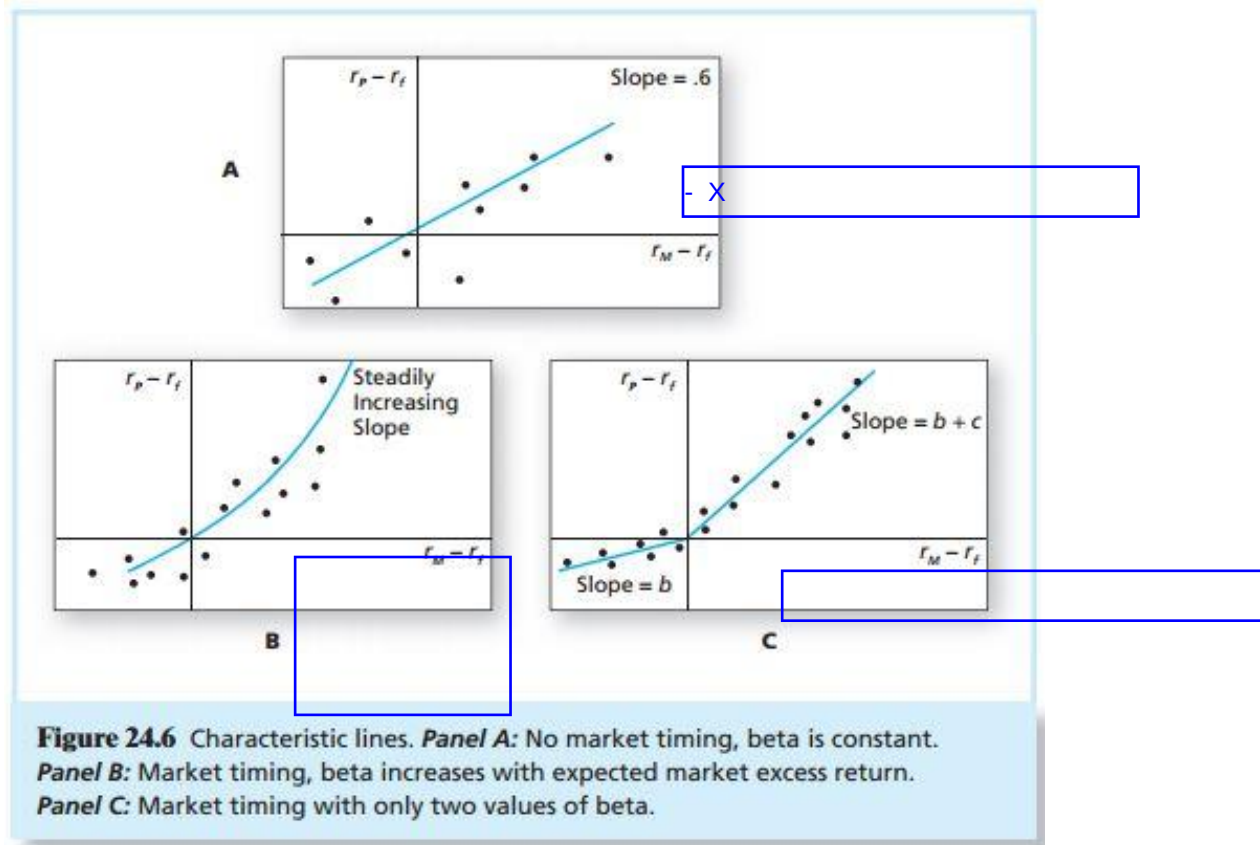
$$r_P - r_f = \boxed{a} + \boxed{b}(r_M - r_f) + c(r_M - r_f)^{\boxed{2}} + e_P$$

- Henriksson and Merton:

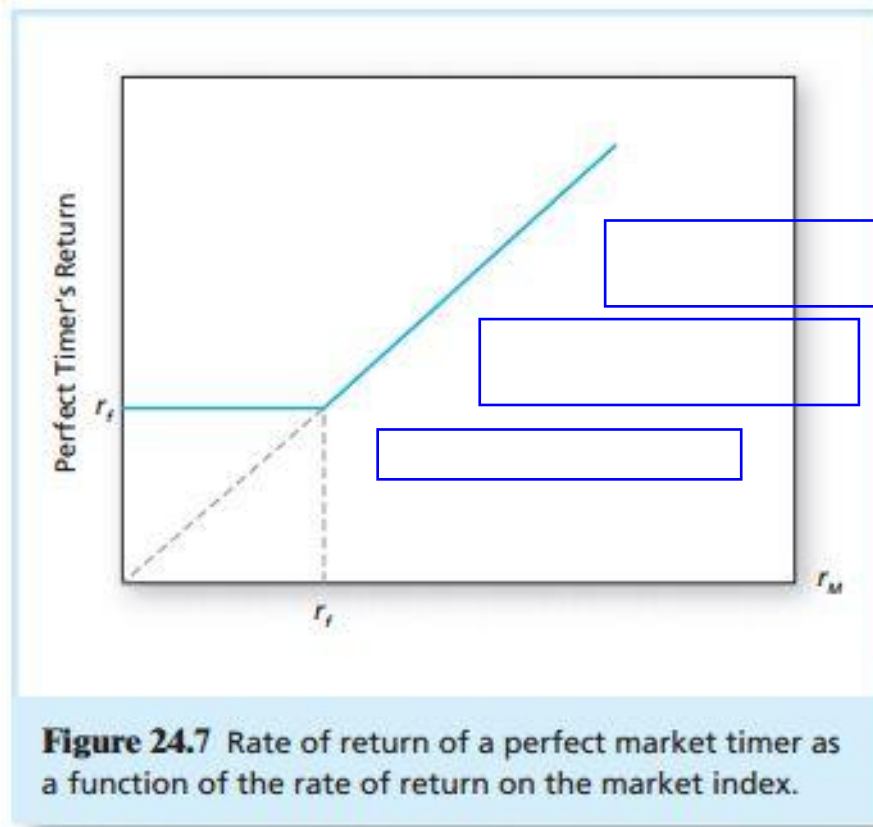
$$r_P - r_f = a + b(r_M - r_f) + c(\boxed{r_M - r_f})D + e_P$$

: binomial;

No Market Timing; Beta Increases with Expected Market Excess Return; Market Timing with Only Two Values of Beta



Rate of Return of a Perfect Market Timer



Performance Attribution

A common attribution system decomposes performance into three components:

1. Allocation choices across broad asset classes.
2. Industry or sector choice within each market.
3. Security choice within each sector.

Attributing Performance to Components

Set up a '**Benchmark**' or '**Bogey**' portfolio:

- Select a benchmark index portfolio for each asset class.
- Choose weights based on market expectations.
- Choose a portfolio of securities within each class by security analysis.

Attributing Performance to Components

- Calculate the return on the 'Bogey' and on the managed portfolio.
- Explain the difference in return based on component weights or selection.
- Summarize the performance differences into appropriate categories.

Formulas for Attribution

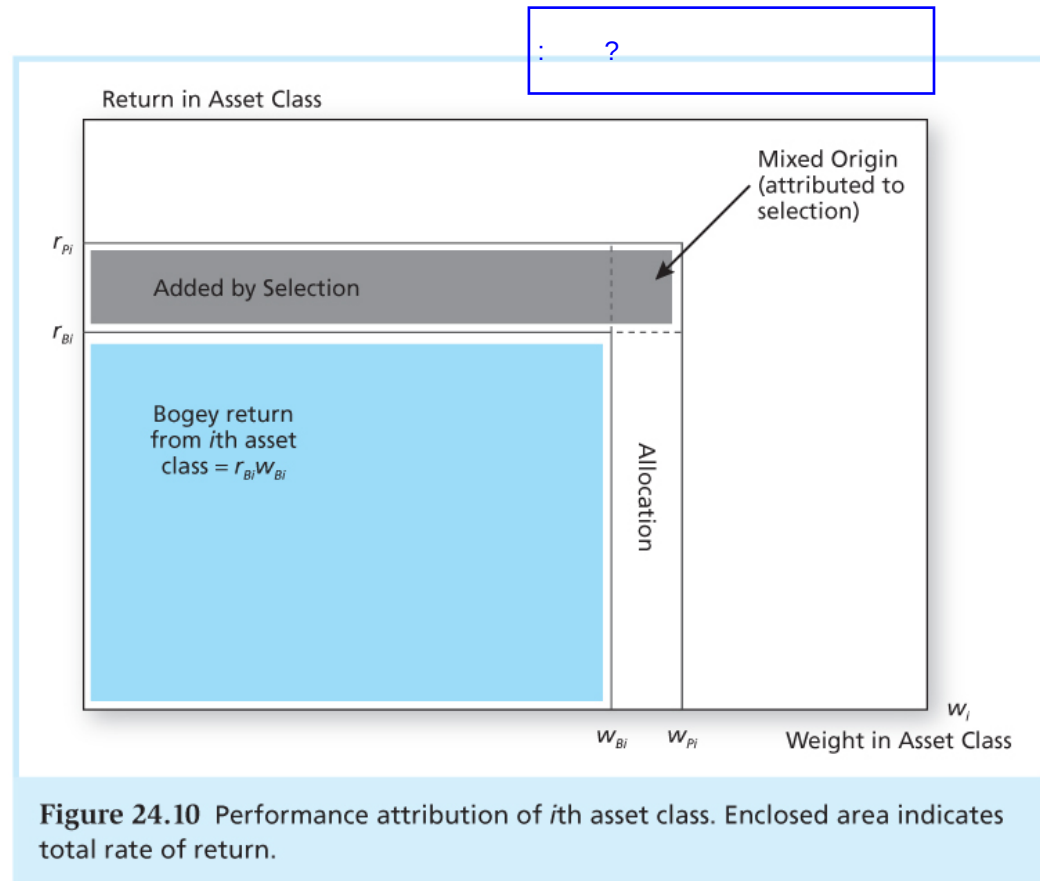
$$r_B = \sum_{i=1}^n w_{Bi} r_{Bi} \quad \& \quad r_p = \sum_{i=1}^n w_{pi} r_{pi}$$

$$r_p - r_B = \sum_{i=1}^n w_{pi} r_{pi} - \sum_{i=1}^n w_{Bi} r_{Bi} =$$

$$\sum_{i=1}^n (w_{pi} r_{pi} - w_{Bi} r_{Bi})$$

Where **B** is the **bogey** portfolio and **p** is the **managed** portfolio

Performance Attribution of *i*th Asset Class



Performance Attribution

Superior performance is achieved by:

- overweighting assets in markets that perform well
- underweighting assets in poorly performing markets

Performance Attribution

Table 24.7

Performance
attribution

A. Contribution of Asset Allocation to Performance					
	(1)	(2)	(3)	(4)	(5) = (3) × (4)
Market	Actual Weight in Market	Benchmark Weight in Market	Active or Excess Weight	Market Return (%)	Contribution to Performance (%)
Equity	.70	.60	.10	5.81	.5810
Fixed-income	.07	.30	-.23	1.45	-.3335
Cash	.23	.10	.13	.48	.0624
Contribution of asset allocation					.3099
B. Contribution of Selection to Total Performance					
	(1)	(2)	(3)	(4)	(5) = (3) × (4)
Market	Portfolio Performance (%)	Index Performance (%)	Excess Performance (%)	Portfolio Weight	Contribution (%)
Equity	7.28	5.81	1.47	.70	1.03
Fixed-income	1.89	1.45	0.44	.07	0.03
Contribution of selection within markets					1.06

Sector and Security Selection

- Good performance (a positive contribution) derives from overweighting high-performing sectors
- Good performance also derives from underweighting poorly performing sectors.